

An Overview of Reinforcement Learning Algorithms for Causal Discovery

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Abstract This chapter explores the integration of reinforcement learning (RL) and causal discovery, providing an in-depth overview of methods that leverage RL for identifying causal structures in complex environments. We begin with a discussion of foundational concepts in causal discovery, including directed acyclic graphs, structural equation models, and interventions, as well as key principles of reinforcement learning. The core of this chapter categorizes and reviews RL-based causal discovery methods into offline and online approaches. Offline methods focus on optimizing causal structures using pre-collected data, while online methods dynamically interact with the environment, performing interventions to refine both causal models and policies. The taxonomy of approaches highlights strategies such as score-based optimization, ordering-based frameworks, and meta-learning. The chapter concludes with a discussion of challenges and future directions, underscoring the transformative impact of combining causal reasoning with reinforcement learning for advancing explainable and adaptive AI systems.

1 Introduction

Understanding causal relationships is a necessary requirement for the safe and robust application of machine learning algorithms in numerous scientific and engineering disciplines [14]. Agents acting in an unknown environment can use a structural causal model to make accurate predictions and inform the decision-making process.

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Furthermore, the use of such causal models is foundational to developing robust explanations of the agent’s behavior.

Causal discovery, the process of uncovering cause-and-effect relationships from observational or experimental data, is fundamental in environments for which the causal model is unknown. However, traditional causal discovery methods frequently encounter significant challenges when dealing with high-dimensional data and intricate causal dependencies [23]. Such limitations restrict their applicability in real-world contexts, where data is often both abundant and complex.

Recent advances in reinforcement learning (RL) offer new opportunities to overcome these limitations. Especially deep reinforcement learning (DRL) has become the state-of-the-art algorithm class for solving sequential decision-making problems. As applications to other domains such as game-playing [20, 22] and robotics [38] have shown, DRL methods are particularly effective in navigating high-dimensional state spaces. By framing causal discovery as a reinforcement learning problem, we can leverage the strengths of RL to identify causal structures in complex data settings.

Causal discovery through reinforcement learning involves using RL techniques to explore and learn the causal relationships between variables in an environment. This approach treats the identification of causal structures as an optimization problem, where the RL agent interacts with the environment, collects data, and uses this information to infer or redirect causal links. Unlike traditional methods that rely on purely statistical dependencies, RL-based approaches can actively intervene in the environment, thereby testing hypotheses and refining causal models in a data-driven manner. By understanding which variables are likely to influence one another, agents can design their actions to test specific causal hypotheses, thereby reducing the number of samples needed to identify the causal structure. This efficiency is particularly important in domains where data collection is expensive or time-consuming, such as in medical research or controlled scientific experiments.

In this chapter, we will provide an overview of methods for causal discovery through reinforcement learning, highlighting both the computational techniques and theoretical advancements enabled by this integration. In **Section 2**, we will begin by examining the foundational concepts of causal discovery, including an introduction to causality, interventions, and causal graphical models. Additionally, we will introduce the core principles of reinforcement learning that are necessary to understand its role in causal discovery. **Section 3** explores the intersection of causal discovery and reinforcement learning (RL), referencing a taxonomy by Méndez-Molina et al. [19] that categorizes methods leveraging RL for causal inference tasks. These methods are classified into three categories: using causal models to enhance RL ($CM \rightarrow RL$), employing RL to learn causal relationships ($RL \rightarrow CM$), and integrating both causal learning and policy optimization ($RL \leftrightarrow CM$). The given chapter focuses on the latter two categories, providing detailed analyses of algorithms that either use RL to discover causal structures from observational or interventional data or simultaneously learn causal relationships and optimize policies. The section begins with methods for offline RL applied to causal discovery, such as score-based optimization techniques, tree search frameworks, and ordering-based strategies. Subsequently, the section delves

into online RL methods, which actively perform interventions in the environment to refine both the causal model and the agent’s policy. Through these sections, we aim to provide a comprehensive understanding of how reinforcement learning can be effectively applied to causal discovery, addressing complex problems in dynamic and uncertain environments. Finally, in **Section 4**, we reflect on the key insights gained and discuss potential directions for future research to further advance the integration of reinforcement learning and causal discovery.

2 Preliminaries

2.1 Causal Discovery Fundamentals

A causal relationship refers to the change in one variable (the cause) producing a change in another variable (the effect). Many machine learning models learn based on statistical associations such as correlations. In contrast, causal models try to identify the underlying mechanism that leads to observed outcomes, which allows us to predict and explain the outcomes of complex control systems. Hence, understanding causality is crucial in domains in which AI algorithms could potentially cause harm, such as personalized medicine [31] and self-driving cars [16].

In the following, we will introduce some common notations from the domain of causal discovery. A graphical representation used to model causal relationships among variables is the Directed Acyclic Graph (DAG). Its nodes represent random variables $\{X_1, X_2, \dots, X_n\}$ and its directed edges $X_i \rightarrow X_j$ indicate that X_i is a direct cause of X_j . The acyclic property ensures that there are no feedback loops, meaning no sequence of directed edges starts and ends at the same node. This characteristic is essential for establishing clear directional relationships between variables. While causal models do not have to be acyclic in general, it is a feasible assumption considering the dynamics of many systems. In contrast, cyclic causal models [30] allow for the existence of feedback loops among the variables. Since to the best of our knowledge, reinforcement learning has not yet been applied to causal discovery for cyclic causal models, we restrict the remainder of this work to acyclic causal models.

Next to acyclicity, we ensure that each variable in a DAG is conditionally independent of its non-effects given its direct causes (also called parents). This property, known as the Markov Condition [2], is crucial because it provides a compact way of representing the dependencies in the system. Specifically, the Markov Condition implies that each variable is independent of all other variables, except for its effects, given its parents. This condition helps in breaking down complex joint probability distributions into simpler, localized conditional distributions, which makes causal inference computationally feasible.

2.1.1 Structural Equation Models

Structural Equation Models [28] provide a mathematical framework to model causal relationships. Here, each variable is expressed as a structural equation represented by a function of its direct causes and an error term:

$$X_i = f_i(\text{PA}_i, \epsilon_i)$$

where PA_i denotes the set of parent variables (direct causes) of X_i , and ϵ_i is an error term representing unobserved influences, assumed to be independent of other error terms. SEMs provide a basis for causal inference by specifying how interventions propagate through the system.

2.1.2 Interventions and Do-Calculus

Another key concept in causal learning is intervention. To formalize interventions, Pearl’s Do-Calculus [25, 27] provides a set of rules that allow for reasoning about how probability distributions change under interventions. Intervening on a variable, often denoted as $do(X = x)$, refers to actively setting the value of a variable rather than observing it passively. Interventions are a core aspect of causal discovery because they allow us to identify the direction of causality. For instance, by intervening on X and observing changes in Y , we can infer whether X is a cause of Y . This kind of reasoning goes beyond correlation and allows for the determination of causal effects.

To assess causal relationships through experimentation, we could test variable A’s impact on another variable B by direct intervention on A and observation of the outcome of B. This process is used for example in medical randomized controlled trials [36], where eligible patients are first randomly grouped into two groups, in which one group is given medication, whereas the other serves as the control group and instead receives a placebo.

While randomized controlled trials are the de-facto gold standard in medicine, they cannot always be used to identify cause-and-effect relationships, because either intervening or not intervening could potentially be too dangerous for the patients. Luckily, such relationships can also be derived from observational data. Following Reichenbach’s Common Cause principle [29] observing two statistically dependent variables implies that there must exist a causal explanation for this dependence. Specifically, either one variable is a direct cause of the other, or there exists a third variable that acts as a common cause for both. Reichenbach’s principle provides the basis for distinguishing between mere correlation and causation by focusing on the underlying causal mechanisms.

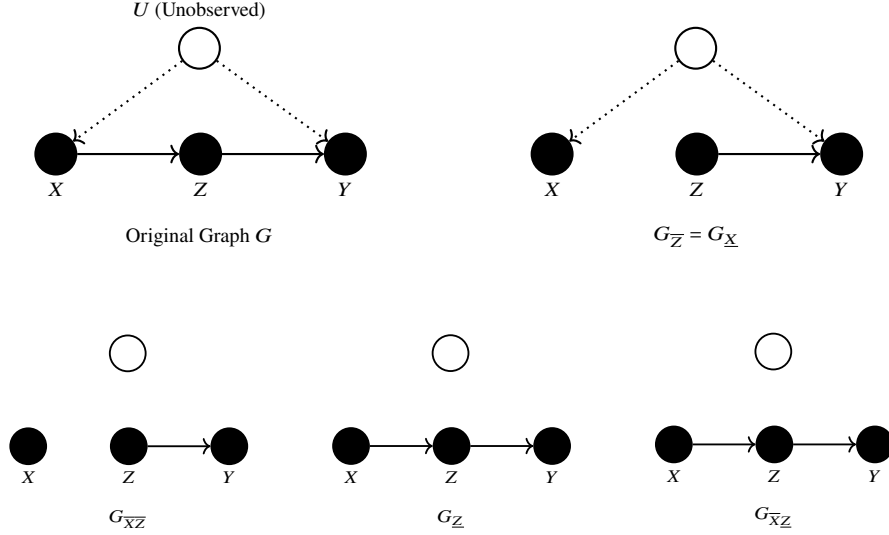


Fig. 1 Graph perturbations used for Pearl's Do Calculus [26].

2.1.3 Pearl's Do-Calculus

Pearl's Do-Calculus [25, 27] is particularly useful when it comes to estimating causal effects from observational data, especially when controlled experiments are not feasible. The application of these rules helps in transforming probabilistic statements about observations into statements about interventions.

Do-Calculus is based on three inference rules that specify how causal relationships can be determined from purely observational data. Let X, Y, Z , and W be arbitrary disjoint sets of nodes in a causal DAG G . The graph obtained by deleting all arrows pointing to nodes in X will be denoted by $G_{\overline{X}}$. Vice-versa, the graph obtained from removing all arrows emerging from nodes in X is denoted by $G_{\underline{X}}$. Various modifications of DAGs are visualized in Figure 1.

1. Rule 1: Insertion / Deletion of Observations

$$P(y|do(x), z, w) = P(y|do(x), w) \quad \text{if } (Y \perp\!\!\!\perp Z|X, W) \quad \text{in } G_{\overline{X}} \quad (1)$$

This rule allows for the insertion or deletion of observations of certain variables, given that these variables are conditionally independent of others, given an intervention.

2. Rule 2: Action/Observation exchange

$$P(y|do(x), do(z), w) = P(y|do(x), z, w) \quad \text{if } (Y \perp\!\!\!\perp Z|X, W) \quad \text{in } G_{\overline{XZ}} \quad (2)$$

This rule provides conditions under which we can replace an intervention with an observation, or vice versa. It allows for the exchange of actions (interventions) and observations, simplifying the identification of causal effects by converting interventions into equivalent observational relationships.

3. Rule 3: Insertion / Deletion of Action

$$P(y|do(x), do(z), w) = P(y|do(x), w) \quad \text{if} \quad (Y \perp\!\!\!\perp Z|X, W) \quad \text{in} \quad G_{\overline{X}, \overline{Z(W)}} \quad (3)$$

This rule allows us to add or remove interventions from the causal graph if these interventions do not affect the variables of interest, given certain conditions. This rule helps reduce the complexity of the causal graph by eliminating redundant interventions and making causal inference more tractable.

2.2 Traditional Methods for Causal Discovery

Causal discovery methods aim to infer the causal DAG from data. They can be categorized into three main types: constraint-based, score-based, and hybrid methods.

Constraint-based methods use statistical tests to infer conditional independence and build a causal graph consistent with the observed data. Being one of the first algorithms published for causal discovery, the PC algorithm [35] as introduced by Peter Spirtes and Clark Glymour starts with a fully connected undirected graph. To identify the causal graph's skeleton, we stepwise remove edges between variables that are conditionally independent given some conditioning set S . This process results in an undirected graph in which the edges need to be oriented such that there are no cycles. Due to possible degrees of freedom in the edge's orientation, this process results in any DAG of the Markov equivalence class of graphs given the provided data set.

Score-Based Methods aim to find the DAG that optimizes a predefined scoring function

$$\min_G \text{Score}(G), \text{ subject to } G \in \text{DAGs}. \quad (4)$$

Common scoring functions such as the Bayesian Information Criterion (BIC) [33], Minimum Description Length (MDL) [5], and Bayesian Gaussian equivalent (BGe) [11] balance the model's fit and its complexity. Starting with any given graph, the structure is optimized by adding, deleting, or reversing edges to improve the score. For a small number of variables, dynamic programming can be used to retrieve the optimal graph structure. Due to the super-exponential number of possible DAGs [4, 6], finding the global optimum quickly becomes too expensive. For this reason, greedy search or heuristic methods are frequently used to find a local optimum [1].

Hybrid Methods combine constraint-based and score-based approaches to leverage their respective strengths. Exemplary for this method class are the two phases of the Max-Min Hill Climbing method [39]. First, a skeleton is identified using

conditional independence tests. Second, a greedy score-based search restricted by the identified skeleton is used. The first phase results in a reduced search space for the score-based optimization, thus improving the efficiency of the optimization.

Over the years, many algorithms have been developed following these common principles. Since our work focuses on causal discovery using reinforcement learning methods, we will introduce related principles next. For readers who would like to read more on causal discovery methods in general we recommend more detailed surveys on that matter [46, 24].

2.3 Reinforcement Learning

Reinforcement learning describes the process of an agent learning to make decisions based on continuous interactions with a given environment to maximize the cumulative reward. RL algorithms are particularly suited for sequential decision-making tasks and have been successfully applied in various fields such as robotics [38], game-playing [22], and autonomous systems [13].

The interaction between the agent and the environment is typically framed as a Markov Decision Process (MDP) [37], defined by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where:

- $\mathcal{S}$ is the set of states.
- $\mathcal{A}$ is the set of actions.
- $P(s'|s, a)$ is the state transition probability, representing the probability of moving to state s' from state s after taking action a .
- $R(s, a, s')$ is the reward function, giving the immediate reward received after transitioning from s to s' via action a .
- $\gamma \in [0, 1]$ is the discount factor, which balances the importance of immediate and future rewards.

At each discrete time step t , the agent observes the current state $S_t \in \mathcal{S}$, chooses an action $A_t \in \mathcal{A}$, based on which the environment transitions to the next state S_{t+1} according to $P(S_{t+1}|S_t, A_t)$ and potentially issues a reward $R_{t+1} = R(S_t, A_t, S_{t+1})$. The standard interaction process is depicted in Figure 2.

The agent's objective is to maximize the expected cumulative reward, also known as the return:

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \quad (5)$$

where γ is a discount factor ensuring that the sum converges when rewards are bounded and gives preference to immediate rewards over distant future rewards. To maximize the return, the agent may learn a policy π , which maps any given state to a probability distribution over actions. The optimal policy, which maximizes G from each state is denoted by π^* .

Once a policy is fixed, we can determine the expected return when starting from a state s and following the given policy.

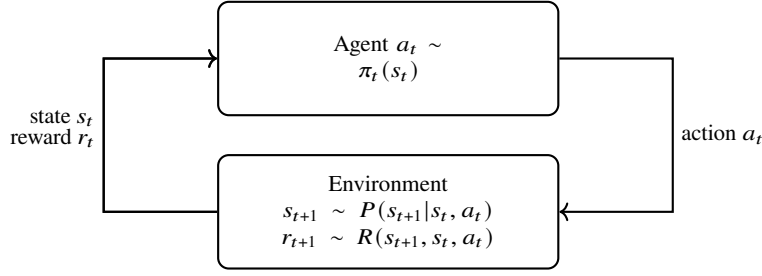


Fig. 2 Interaction loop in reinforcement learning [37].

$$V^\pi(s) = \mathbf{E}_\pi[G_t | S_t = s] \quad (6)$$

We call $V^\pi(s)$ the (state-)value function. Similarly, the expected return when starting from state s , taking action a , and thereafter following policy π is called the action value function and is denoted by

$$Q^\pi(s, a) = \mathbf{E}_\pi[G_t | S_t = s] \quad (7)$$

The optimal state-value function $V^*(s)$ and the optimal action-value function $Q^*(s, a)$ are defined as the maximum expected return achievable from state s or state-action pair.

2.3.1 Types of Reinforcement Learning Algorithms

RL algorithms are commonly classified according to their learning target. Typically, this results in the categories: value-based, policy-based, and actor-critic methods.

Value-based methods focus on estimating the state-value function $V(s)$ or the action-value function $Q(s, a)$. Policies are indirectly derived from these value functions by taking the best possible action for the given state. Q-learning [42] compares its current estimate of the action value function with recent observations and updates it accordingly. The concept has later been extended by deep Q networks (DQN) [22] in which deep neural networks are used to approximate the values of the action value function.

Policy-Based methods such as the Reinforce algorithm [43] directly optimize the parameters of the policy function to maximize the expected return. Similarly to DQN, the policy function is typically represented by a neural network.

Actor-critic methods combine value-based and policy-based methods by maintaining both a policy (actor) and a value function (critic). Methods such as Advantage Actor-Critic [21] and (Deep) Deterministic Policy Gradient [34, 15] use gradients provided by the critic to update the actor's policy.

In the following section, we will highlight how these different types of methods can be used for causal discovery and how the respective problem can be encoded as a Markov decision process.

3 Causal Discovery through Reinforcement Learning

Causal learning and reinforcement learning can be combined in multiple ways. Méndez-Molina et al. [19] characterized existing works into three groups:

- **Causal Models to Improve RL ($\text{CM} \rightarrow \text{RL}$):** These works use known causal models to enhance RL agents' action selection, speeding up learning.
- **RL to Learn Causal Relationships ($\text{RL} \rightarrow \text{CM}$):** In this approach, RL is either used to learn causal relationships through interventions or learns a policy to optimize the structure based on given observational data.
- **Simultaneous Causal Learning and Policy Optimization ($\text{RL} \leftrightarrow \text{CM}$):** Recent works attempt to learn causal relationships and optimize the agent's policy at the same time.

In this work, we will be concentrating on learning causal relationships using RL in the sense of ($\text{RL} \rightarrow \text{CM}$) or ($\text{RL} \leftrightarrow \text{CM}$). For the latter, we specifically focus on the strategies for learning a causal structure and the advantages or disadvantages of doing so in an intertwined manner. This way we aim to isolate the task of causal discovery from strategies for policy optimization. In some applications, the policy optimization aspect is either already in place or can be handled separately, while causal learning is what's lacking. For instance, in healthcare, policies for treatment might be based on existing medical guidelines, but learning new causal insights from patient data could reveal unexpected causal pathways that might improve these guidelines. Once causal relationships are discovered, they can be transferred and used to inform different policies in other related tasks, without needing to re-learn causality. Similarly, learned causal models can be used outside of RL altogether. Readers interested in using causal models to improve RL are referred to the survey by Zeng et al. [47], which covers a wide range of algorithms in this category.

In the following, we structure our discussion of reinforcement learning methods for causal discovery around the type of data interaction: whether the data is fixed and observational or can be actively collected through interventions. We begin by exploring offline reinforcement learning, where causal discovery is performed using a static dataset, typically without the ability to generate new data. These methods focus on optimizing graph structures or orderings using pre-collected data. Following this, we delve into online reinforcement learning, where causal discovery and policy learning are intertwined. In this setting, agents can actively interact with the environment to perform interventions and iteratively refine their causal models. This dual capability highlights the potential of reinforcement learning to address challenges in both causal inference and decision-making simultaneously. An overview of all discussed methods and their causal discovery approaches is presented in Table 1.

Table 1 Comparison of Reinforcement Learning Methods for Causal Discovery

Method	Authors	RL Type	Causal Discovery Approach
RL-BIC	Zhu et al. [49]	Offline	Score-based optimization of the adjacency matrix
KCRL	Hasan et al. [12]	Offline	Prior knowledge-guided score-based optimization of the adjacency matrix
CD-UCT	Darvariu et al. [8]	Offline	Score-based tree search in DAG space
CORL	Wang et al. [41]	Offline	Score-based optimization of variable orderings
RCL-OG	Yang et al. [45]	Offline	Score-based optimization of samples from a learned posterior-distribution of variable orderings
Causal Q-learning	Méndez-Molina [18]	Online	Interventional causal discovery
GRADER	Ding et al. [9]	Online	Iterative probabilistic inference
MCD	Sauter et al. [32]	Online	Meta-learning for causal discovery

3.1 Offline Reinforcement Learning for Causal Discovery

For a long time, causal discovery algorithms focused on learning the causal structure from a given set of observations. While RL is often applied in interactive scenarios, such that new observations are continuously added to the data set, offline RL algorithms focus on learning an optimal policy from a given data set. For this reason, algorithms in this subsection fall into the category of $(\text{RL} \rightarrow \text{CM})$.

Zhu et al. [49] use an actor-critic reinforcement learning algorithm for score-based causal discovery. Their algorithm called RL-BIC involves an encoder-decoder neural network model to generate graph adjacency matrices of directed graphs. Here, a transformer architecture [40] is used for the encoder to process the input data. The decoder generates a graph adjacency matrix in an element-wise manner, sampling edges probabilistically using Bernoulli distributions. The RL agent’s objective is not to learn a policy but rather to explore the graph space and select the DAG with the highest reward. The reward function integrates a predefined score function (like BIC) and two penalty terms to enforce acyclicity [48], which is critical to ensuring that the graph is a valid DAG. The reward structure guides the RL model to efficiently search for high-scoring, acyclic graphs. Each action corresponds to adding, removing, reversing, or maintaining directed edges between nodes to form a new graph configuration. During training, the agent adjusts the likelihood of adding or removing edges based on the rewards received. The final output is represented by the graph that maximizes the score among all graphs generated during training.

For cases in which expert knowledge is available, Hasan et al. [12] proposed a framework for Knowledge-based causal discovery (KCRL). It leverages a similar score-based reinforcement learning-based search process, utilizing a self-attention-based encoder-decoder architecture to generate graph adjacency matrices from observational data. A comparator evaluates these matrices against prior knowledge, imposing penalties for mismatches while rewarding adherence. This dual feedback mechanism not only enforces constraints but also accelerates convergence by reducing the feasible graph search space.

Darvariu et al. [8] propose CD-UCT, a model-based reinforcement learning algorithm for causal discovery that uses tree search in DAG space. Unlike RL-BIC, CD-UCT incrementally constructs a causal graph by adding edges one at a time while avoiding cycles using a provably efficient algorithm. The introduction of an incremental cycle detection mechanism enables deeper and more efficient exploration in DAG space. This tree search framework incorporates a Markov Decision Process (MDP) formulation where each state represents the current graph structure, and actions correspond to valid edge additions. The method relies on Upper Confidence Bound for Trees (UCT) to balance exploration and exploitation during graph construction, ensuring the search process identifies high-quality structures.

Wang et al. [41] introduced an alternative optimization-based approach called Causal discovery with Ordering-based Reinforcement Learning (CORL). It integrates reinforcement learning (RL) into an ordering-based framework for identifying causal relationships among variables. Unlike previously discussed methods that directly search through directed acyclic graphs (DAGs), CORL focuses on discovering an optimal variable ordering, significantly reducing the search space. Each action of their encoder-decoder-based architecture represents the next variable to be appended to the current variable ordering. Once an ordering is determined, variable selection methods are applied to derive the final DAG. The reward for any given variable ordering reflects the fit to the observed data’s causal structure. CORL improves scalability and performance compared to previous RL-based methods, demonstrating effectiveness on synthetic and real datasets, even for larger-scale problems.

Yang et al. [45] also leveraged the smaller search space of an ordering-based optimization. Instead of optimizing the score of a single source graph, the proposed RCL-OG method learns the posterior distribution of orderings given the observed data, basing this on the fact that the true DAG may not be identifiable in some settings. The resulting probabilistic model can be used for sampling orderings by traversing the ordering graph. Each action corresponds to adding a variable to a sorted set. Once the set consists of all variables, the completed ordering is used to learn a structure and calculate a reward using BIC. The method uses a symmetric learning architecture that leverages transformers both for the encoder and the decoder. The output of the decoder is passed in input to an MLP to obtain estimates of the Q-values for each action (i.e., the next node to include in the ordering). Authors show that the correct posterior is accurately recovered in some simple examples and that better or comparable performance is obtained on a series of synthetic and real-world benchmarks.

In a reinforcement learning setting, we typically distinguish the agent and the environment as two separate entities. Madumal et al. [17] proposed a causal learning approach that receives causal relationships of the environment variables and a reinforcement learner’s observations and extends it by adding structural equations in the form of a multivariate regression model. The generated model allows users to describe the effects that specific actions have on the environment’s state and therefore may serve as an explanation of the agent’s behavior. While strictly speaking this is not a reinforcement learning algorithm for causal discovery, the reinforcement learning agent proposed by Madumal et al. serves the purpose of generating data for identifying a plausible causal model of the environment. In the following, we will see how these two phases can be combined in a more effective manner.

3.2 Online Reinforcement Learning for Causal Discovery

This second set of approaches leverages RL to perform strategic interventions in an environment, collecting both observational and interventional data to construct Directed Acyclic Graphs (DAGs) representing causal relationships among variables. In contrast to offline reinforcement learning, the current policy determines the observations being collected. For this reason, the optimization of the agent’s causal model of its environment and its policy are often intertwined, falling into the category of (RL $\leftrightarrow$ CM) algorithms.

Méndez-Molina [18] discussed combining both phases to unlock the potential to accelerate causal discovery via interventions. Instead of strictly splitting the task of causal discovery and learning to interact with an environment, the agent iteratively interacts with the environment to collect its own set of observations. Given the growing data set, the agent aims to improve on its causal model of the environment. The resulting process, denominated Causal Q-learning, is repeated until both, the agent’s policy and the causal model, converge.

Ding et al. [9] introduced GRADER (GeneRALizing by DiscoverERing), a reinforcement learning framework that tightly integrates policy optimization with causal discovery to enhance generalization capabilities in Goal-Conditioned Reinforcement Learning. GRADER treats the causal graph as a latent variable within a probabilistic inference framework, iteratively improving both the learned causal structure and the agent’s policy through alternating updates. The RL agent explores the environment and performs tasks while dynamically refining its causal understanding, thereby creating a cycle between causal graph discovery, transition model learning, and policy optimization. To discover the causal graph’s structure GRADER calculates the likelihood that an edge exists using statistical tests (e.g., χ^2 -test for discrete variables or Fast Conditional Independence Test [3] for continuous variables). If two nodes are dependent, a directed edge is added based on predefined causal assumptions. Once the structure of a causal graph has been obtained, a structural equation model matching the agent’s observations is approximated using a neural network. The network consists

of multiple layers in which each variable is represented by gated recurrent units [7] that receive the values of the current variable’s set of parent nodes as input.

Sauter et al. [32] proposed a meta-learning approach called Meta-Causal Discovery (MCD), which aims to integrate observational and interventional data into an RL-based structure learning algorithm. The whole model is trained to learn a policy for generating a causal structure based on given observations and possible interventions. It uses an actor-critic model involving LSTM layers to approximate the history of observations. The RL agent can perform interventions, or “listening actions,” on specific variables to distinguish causal effects. These actions replace the usual dependency of variables on others, simplifying the identification of direct causal connections. Additionally, the agent has “structure actions” to adjust edges in the causal graph, allowing it to add, remove, or reverse edges based on observed data. By training on environments with known causal graphs, MCD has shown to generalize to estimate causal structures in new, unseen settings. Given the limitations of the policy’s horizon, the training is aimed at minimizing the number of interventions required.

4 Conclusion

The integration of causal discovery and reinforcement learning represents a significant step toward more intelligent and autonomous systems. By leveraging RL techniques for causal discovery, we can tackle complex, high-dimensional problems that traditional methods struggle with. Conversely, embedding causal reasoning into RL agents enhances their ability to learn efficiently and adapt to new environments.

However, several open challenges and potential directions remain to be explored. Adjacency matrix-based RL-based causal discovery methods still face computational limitations when dealing with complex causal structures. The development of more scalable algorithms is essential to broaden their applicability. Work on order-based optimization algorithms has already shown great promise in improving scalability by introducing a two-phase optimization scheme. Further development of multi-stage optimization schemes may yield improvements in the scalability of RL-based causal discovery algorithms. Similarly, abstractions of the state and action space may improve the sampling efficiency of RL-based optimization algorithms [10, 44].

Another significant limitation is the reliance on the assumption of acyclic causal structures, which may not align with real-world systems where feedback loops are common. Besides, real-world environments are often dynamic, with causal relationships evolving over time. In principle, RL-based methods are well suited to interacting with and learning from dynamic environments. The development of adaptive and continuous learning approaches for RL-based causal discovery could fill this gap by enabling systems to learn and adapt to changes without forgetting previously acquired knowledge. We believe that RL-based causal discovery algorithms could become a natural fit in environments with non-stationary dynamics, such as climate systems or robotic agents operating in changing conditions.

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